

Exercises for Lecture "Theoretical Spectroscopy for systems implying periodic boundary conditions"

Exercise 3: UV/VIS spectroscopy

based on exercises taken from the [CP2K Homepage](#).

2.1: Absorption spectra, excited-state optimization and emission spectra

- Start by computing an optimized ground-state structure (if possible also checking that the structure is a local minimum structure performing a frequency analysis.) Choose either a semi-empirical method or GGA or hybrid density density functional theory for the computational setup. Note that semi-empirical ground-state computations can only be combined with a corresponding subsequent semi-empirical excited-state treatment. For hybrid functionals you should use the Auxiliary density matrix method with an exemplary input given in `admm_absorption.inp`.
- Based on the optimized ground-state structure, compute the absorption spectrum for the 10 lowest excited states using the Tamm-Dancoff approximation as part of the TDDFT code in CP2K. Note that the computation of excitation energies is switched on by adding the `PROPERTIES` subsection in the `TDDFT` section. The keyword `NSTATES` enables to specify the number of excitation energies to be computed and the keyword `KERNEL` with options `FULL` and `STDA` enables to switch between density functional theory kernels as well as semi-empirical analogues (`STDA` = simplified Tamm-Dancoff approximation). Analyze the output file by taking a look at obtained excitation energies, oscillator strengths and a corresponding analysis of the transitions with respect to the molecular orbital contributions.
- Choose the excited state with the highest oscillator strength (this depends on the chosen computational setup) and optimize the state by adding the subsection `EXCITED_STATES` to the `DFT` section.
- Write a small script to grep absorption and emission wavelengths from the output files and to thereon based compute the Stoke's shift in nm with the Stoke's shift being given as the wavelength of absorption minus the wavelength of emission.